Aashish Ranjan

Roll No.: B24EDP013
B.Sc.- B.Ed. (ITEP)in Physics.
Department of CET (Centre for Education Technology)
Indian Institute of Technology, Jodhpur

+91-9341251390
aashishranjan2601@gmail.com
b24edp013@iitj.ac.in
linkedin.com/in/aashish-ranjan-87472630a/

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Sc.- B.Ed. (ITEP)	Indian Institute of Technology, Jodhpur	7.0	2024 -Present
Senior Secondary	BSEB Board	75.6%	2023
Secondary	CBSE Board	72.6%	2021

EXPERIENCE

- **Physics Teaching Intern.** *The Red Fight Career Solution | 2 Months*
  - Interacted with Class XII students preparing for the JEE Main examination and supported their learning of Physics concepts.
  - Taught the topics of **Physics**, explaining key concepts to help students develop conceptual understanding.
  - Guided students through the practice and solving of **JEE Main Previous Year Questions (PYQs)** related to Physics.
  - Gained experience in explaining technical concepts, engaging with students, and supporting exam-oriented Physics learning.
  - **Teaching-Learning Resources:** Board and marker, digital board, PhET Interactive Simulations.

PROJECTS

- **Development of a Custom-Made Tunable Optical Spectrometer.**
  - Explored the design of a compact optical spectrometer using diffraction principles and camera-based spectral recording.
  - Investigated the role of optical slits, diffraction, and interference in spectrum formation and resolution.
  - Worked on capturing and analysing spectra, including intensity-versus-wavelength plots and comparisons of measurements with and without a slit.
  - **Tools & technologies used**: Optical slits, CD/DVD diffraction elements, camera-based spectral recording, Holmarc Image Analyser Pro.
- **COMSOL-Based Simulation of the Otto Configuration for Surface Plasmon Resonance**
  - Explored the Otto configuration for optical coupling and surface plasmon resonance in plasmonic sensing systems.
  - Studied total internal reflection, evanescent fields, TM polarisation, and electromagnetic boundary conditions.
  - Worked on developing a computational model to investigate optical behaviour in the Otto configuration.
  - **Tools and Technologies used**: COMSOL Multiphysics, electromagnetic theory, plasmonic.
- **Marble Graviton: Game-Based Learning for Physics Education.**
  - Designed a physics-based educational game concept for school students in Classes 9–12 to promote experiential and activity-based learning.
  - Integrated concepts of circular motion, centripetal force, gravitational force, normal reaction, and friction into a game-based challenge.
  - Developed a learning approach encouraging students to predict outcomes, perform challenges, observe results, and explain physical phenomena.
  - **Focus Areas**: Game-based learning, physics pedagogy, experiential learning, instructional design.

CERTIFICATION COURSE

Node Js: Codecademy
Android Development: Internshala
Git & GitHub: Codecademy
Public Relation & communication Management: Udemy

TECHNICAL SKILLS

- **Programming:** Python, Frontend Development, Android Development
- **Simulation Tools:** COMSOLE Multiphysics,

POSITIONS OF RESPONSIBILITY

- Secretary, RAW Media & outreach Society, IIT Jodhpur — Aug. 2025- Aug. 2026
- Core Member, Shutterbugs, IIT Jodhpur — Aug. 2025- Aug. 2026
- Assistant Head, IGNUS, IIT Jodhpur — Mar. 2026
- Assistant Head, Sandstone, IIT Jodhpur — Sept.2025
- Volunteer, International Relation Conclave, IIT Jodhpur — Oct.2025